TANet: An Unsupervised Two-Stream Autoencoder Network for Hyperspectral Unmixing

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***Abstract*— Spectral unmixing is a major technique for the further development of hyperspectral analysis. It aims to determine the corresponding proportion (fractional abundance) of the basic spectral signatures (endmembers) blindly at the subpixel level. Recently, the learning-based method has received much attention in hyperspectral unmixing, and autoencoders have been effectively designed to solve the unsupervised scenarios of unmixing. However, their ability to extract physically meaningful endmembers remains limited, and the performance has not been satisfactory. In this article, we propose a novel two-stream network, termed TANet, to address the above problems. The network consists of a two-stream architecture. First, superpixel segmentation is adopted as preprocessing to extract the endmember bundles from the image. Then, the first stream learns a mapping from the pseudopure pixels to their corresponding abundances. The second stream is conducting the same untied-weighted autoencoder to minimize reconstruction errors from the original pixel data. By learning from the pure or nearly pure candidate pixels to correct the weights of unmixing, the proposed TANet exhibits a more accurate and interpretable unmixing performance. Extensive experiments on both synthetic and real hyperspectral data demonstrate that the proposed TANet can outperform the other state-of-the-art approaches.**

***Index Terms*— Autoencoder, deep learning, hyperspectral image (HSI), hyperspectral unmixing.**

I. Introduction

HYPERSPECTRAL image (HSI) analysis has gained considerable attention due to its rich content information and has played an important role in a wide range of fields, such as military investigation, target tracking, object recognition, and environment monitoring [1]–[5]. However, due to the relatively low spatial resolution of instruments and complex natural surfaces, a pixel in HSI always contains more than one spectral signature, which leads to the "mixed pixels" phenomenon [6], [7]. The existence of a large number of mixed pixels has a serious impact on the accurate recognition of image content and hinders the application and development of the HSI analysis. In this case, extracting basic features (i.e., endmembers) at the subpixel level and calculating the corresponding proportion (i.e., fractional abundance) have become the major preprocessing techniques for HSI analysis. This task is also called spectral unmixing (SU).

According to the specific mixing models of the image pixels, the method proposed to solve the SU task can be simplified by the linear mixing model (LMM), which assumes that the physical region corresponding to a pixel contains several pure materials, so that the observed spectra are linearly combined by the endmembers and their respective abundance coefficients. Geometrical- and statistical-based approaches are the mainstream unmixing methods in the past period of time. Geometrical-based is to project the observed pixels into several subspaces and takes the vertices of the simplex as the extracted endmember in their abundance estimation. The vertex component analysis (VCA) [8], N-FINDR [9], and iterative error analysis (IEA) [10] are the representative methods that belong to this category. These methods are based on the assumption of the availability of pure signatures in the image. Nevertheless, the pure pixels assumption may not hold for real scenarios, especially for the data with relatively low spatial resolution. Nonpure-pixel-based methods are proposed for processing the highly mixed HSI data, such as [11]–[14]. Besides, statistical-based approaches are also a powerful alternative to unmix highly mixed data and have drawn much attention. For instance, the Bayesian framework focuses on exploiting the statistical assumptions and imposing priors to constrain the solution space. In [15], the abundance is modeled as mixtures of Dirichlet densities, where the physical constraints of abundance can be naturally incorporated [i.e., the abundance of none-negative constraint (ANC) and the abundance of sum-to-one constraint (ASC)]. In this article, we use conventional algorithms to denote these approaches, which derives estimations from the image itself without learning. As the main trend in the past few decades, the main challenges faced by these algorithms are whether the convergence direction is close to the optimal solution and the speed of convergence. Nevertheless, for these conventional algorithms, since the unmixing task is essentially an optimization problem, each latent variables derived from these algorithms both have explicit and physical significance.

Recently, learning-based and artificial neural network approaches have demonstrated superior performance in many fields of studies, such as computer vision, natural language processing, speech recognition, and many other fields. The researchers also studied the application of deep neural networks in the HSI analysis. Special attention is paid to HSI classification [16]–[18]. However, although the potential of neural networks in solving inverse problems has been recognized,

Manuscript received August 11, 2020; revised December 4, 2020, April 8, 2021, and May 24, 2021; accepted June 9, 2021. Date of publication July 19, 2021; date of current version January 14, 2022. This work was supported by the National Natural Science Foundation of China under Grant 61903279, Grant 61805181, Grant 61773295, and Grant 61601397. *(Corresponding author: Xiaoguang Mei.)*

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Digital Object Identifier 10.1109/TGRS.2021.3094884

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the application of neural networks for addressing unsupervised SU problems is a handful. The approaches in [19] and [20] develop learning-based methods for estimating the abundance fractions and exhibit better performance compared with conventional algorithms. Nevertheless, the endmember signatures should be known in advance, so their practicality is limited. Autoencoder becomes a natural starting point for investigating the unsupervised SU purpose and has achieved rapid development. The autoencoder can be simply described as learning the low-dimensional representation (i.e., abundance fractions) of data by minimizing reconstruction errors [21]–[23]. This is ideally suitable for conducting an autoencoder network to obtain both endmembers and abundance fractions simultaneously. With inheriting the advanced denoising and intrinsic self-adaptation capabilities of the neural network, [24]–[27] exhibit good performance by employing a symmetrical architecture autoencoder for the unmixing task, where the weight matrices of the encoder and the decoder are tied-weighted. However, in many scenarios, the tied-weighted assumption may do not hold and would ultimately lead to ineffective results on endmember extraction [28]. More recently, a few nontied weighted architectures have been proposed and demonstrated the good potential of autoencoders in the task of unmixing. Specifically, in [26], the approaches are designed to concatenate a marginalized denoising autoencoder (MDA) to improve the robustness of the framework. Nevertheless, this kind of non-end-to-end approach will easily introduce additional denoising errors before unmixing. To address the issue of outliers, staked nonnegative sparse autoencoder (SNSA) is employed to obtain good endmembers initialization for the unmixing process [29], [30]. Convolutional autoencoder [31] and nonlinear autoencoder [32] are also developed for dealing with unsupervised unmixing task.

A significant disadvantage of learning-based approaches is that the training procedure is closer to a black-box mechanism, which causes difficulties to characterize the layers to explicitly link in specific semantics. Such disadvantage always leads to the generalization and accuracy heavily depends on the quantity and quality of training samples [33]. When it comes to the specific task of SU, such limitations also hinder the unmixing of performance to be further improved. For the most existing autoencoder-based approaches, the weight matrix of the decoder is directly regarded as the obtained endmember signature, extracting physically meaningful endmembers remains limited. The endmember is defined that the ground area corresponding to a pixel contains only one characteristic feature. Due to the lack of effective guidance for the endmember properties, only by finding the solution with the smallest reconstruction error while satisfying the physical meaning of abundance (ANC and ASC), the network tends to generate trivial and meaningless unmixing results. To address the above problems, some current approaches (i.e., untied denoising autoencoder (uDAS) [28], sparse autoencoder network (EndNet) [27], deep autoencoder network (DAEN) [29], and deep generative endmember model (DeepGUn) [34]) adopt the pure-pixel-based endmember extraction algorithm (i.e., VCA [8]) to initialize weight of the network; thereby, the physical information of the endmember can be incorporated into the network for training. Nevertheless, the instability of endmember extraction limits the unmixing performance to be further improved. Specifically, in [35], a novel share-weighted network called Deep HSNet is proposed for the unmixing task, which introduces the endmember properties based on different algorithms into a half-siamese network and yielding more reasonable and superior unmixing results. However, Deep HSNet is a non-end-to-end structure, which determines the network's stability and performance mainly depending on the accuracy of the extracted endmembers and the abundance initialization. In dealing with more complex scenarios, noise and initialization conditions will greatly affect the estimated solution. Hence, how to take the endmember properties fully into consideration and solve the problem of unstable initialization during the process of unmixing is the main concern of our algorithm.

In this article, we develop an entirely end-to-end network for hyperspectral unmixing, termed TANet. The proposed TANet consists of a two-stream architecture. The first network learns a mapping from the pseudopure pixels extracted from the image to their corresponding abundances. The second network is conducting the same untied-weighted autoencoder to minimize reconstruction errors from the original pixel data. The parameters of the two networks are forced to share with the same weight, making it possible to transfer the intrinsic properties of the endmembers into consideration during the process of unmixing. Compared with other learning-based algorithms, the two-stream network is equivalent to providing an additional adaptive training error to the traditional autoencoders and further improves the robustness and accuracy of the network. The main contributions of our proposed TANet are threefold. First, according to the problems existing in the current autoencoder-based approaches, we propose a two-stream architecture for the SU task, which can solve the issues of lacking effective guidance for the endmember properties and the unstable initialization during the process of unmixing. By learning from the pure or nearly pure candidate pixels to correct the weight of the unmixing network, the proposed TANet can exhibit a more accurate and interpretable unmixing performance. Second, we develop a new and effective strategy for extracting the endmember bundles based on superpixel segmentation, where the parameters of strategy do not need to be carefully adjusted and the spatial information in the scene can be fully utilized. Third, the proposed method can be fully implemented in an unsupervised manner; the meaningful endmember signatures and abundance fractions can be obtained simultaneously.

The rest of this article is structured as follows. The problem formulation related by autoencoder is briefly introduced in Section II. The detailed architecture of the proposed method is described in Section III. The experimental method and results are shown in Section IV. Finally, some concluding remarks are summarized in Section V.

II. Problem Formulation

Since the proposed method is an autoencoder-based framework, the formulation of this network will be introduced in Section II briefly.

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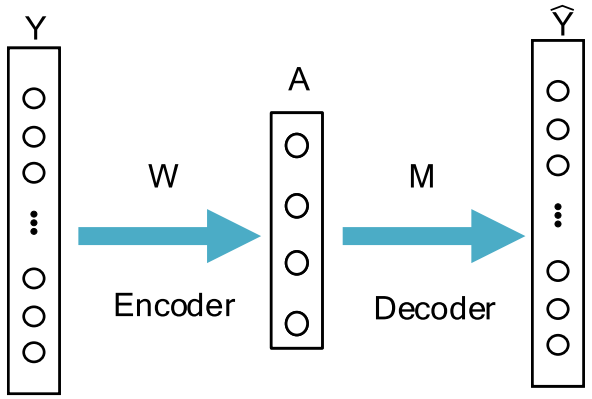


Fig. 1. Flowchart of the traditional autoencoder designed for unmixing scenarios.

Under the LMM assumption, the observed spectra $\boldsymbol{y}_n \in \mathbb{R}^B, n = 1, \ldots, N$ is a linear combination of the spectra $m_j \in \mathbb{R}^B, j = 1, \ldots R$ according to their respective proportion α_{nj}, that is

$$\boldsymbol{y}_n = \sum_{j=1}^{R} \boldsymbol{m}_j \alpha_{\mathrm{nj}} + \boldsymbol{n}_n. \text{ s.t. } \alpha_{\mathrm{nj}} \geq 0, \sum_{j=1}^{R} \alpha_{nj} = 1 \tag{1}$$

where R denotes the number of endmembers, α_{nj} is the corresponding abundance for the jth endmember at the nth pixel, $\alpha_{\mathrm{nj}} \geq 0$, and $\sum_{j=1}^{R} \alpha_{\mathrm{nj}} = 1$ denotes the ANC and ASC constraints, respectively. Autoencoder is a common tool in the artificial neural network, which aims to learn a representation for a set of data, typically for dimensionality reduction. Since the unmixing procedure can be interpreted as finding a set of low-dimensional representations (abundances) and reconstructing data with corresponding bases (endmembers), the interpretation is ideally suited for conducting unsupervised unmixing scenarios. The traditional autoencoder for the unmixing task can be designed into three-layer architecture, which is illustrated in Fig. 1. The function of transforming can be mathematically expressed as

$$\mathbf{A} = f(\mathbf{Y}) = \sigma(\mathbf{W}\mathbf{Y}) \tag{2}$$

$$\hat{\mathbf{Y}} = g(\mathbf{A}) = \mathbf{M}\mathbf{A} \tag{3}$$

where $\mathbf{Y}$ and $\hat{\mathbf{Y}}$ denote the input pixel data and reconstructed pixel data, respectively. The output of the hidden layer $\mathbf{A}$ represents the obtained abundance. $\mathbf{W}$ and $\mathbf{M}$ denote the weight matrix of the encoder and the decoder. Here, $\mathbf{M}$ is the obtained endmembers signature. $\sigma(x)$ represents the adopted activation function used in the network. Given this architecture and formulation, the reconstruction error of the training procedure can be expressed as follows:

$$J(\mathbf{M}, \mathbf{A}) = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{2} \| g(f(\mathbf{y}_i)) - \mathbf{y}_i \|^2. \tag{4}$$

III. Proposed Method

The corresponding architecture flowchart is shown in Fig. 2, which consists of a two-stream architecture, an endmember network (EN), and an unmixing network (UN). $\mathbf{Y} \in \mathbb{R}^B$ denotes the input hyperspectral data, and $\hat{\mathbf{Y}}$ denotes the reconstructed hyperspectral data generated by the decoder. The output of the hidden layer $\mathbf{A}$ represents the estimated abundances, and the weight matrix of the decoder represents the obtained endmember signatures. $\mathbf{A}_{\mathrm{slic}}$ represents the abundance features that are derived from the pure or nearly pure candidate pixels. Note that the sharing strategy is under the assumption that the weight matrix of two networks (EN and UN) has the same meaning since the endmember signatures of two networks are both estimated from the same image. The proposed method mainly consists of two procedures. The first one is adapting an efficient clustering strategy to extract sets of endmember bundles by integrating the spatial and spectral information contained in the observed data. Then, the endmember bundles are gathered together and fed to the EN by minimizing reconstruction errors. On the other hand, the specific layer type of two networks (EN and UN) is designed to be the same structure. The parameters of the decoder of two networks are also forced to share weights. By learning from the candidate pseudopure pixels mapping to their abundance, the EN can be regarded as providing an additional training error to correct the TANet converging toward a more accurate and interpretable unmixing solution. Finally, meaningful endmembers and corresponding abundances can be obtained.

A. Region-Based Endmember Bundles Extraction From the Observed HSI

From the network structure illustrated in Fig. 1, we can see that, for the traditional autoencoder-based algorithms, the weight matrix of the decoder $\mathbf{M}$ is directly interpreted as the obtained endmember signature. Although the network can effectively deal with unsupervised scenarios by incorporating the ANC and ASC constraints into the network, the results are not convincing, and the physical meaning of the endmembers cannot be satisfied. Moreover, the accuracy and robustness are limited. EN is training with pseudopure pixels mapping to their abundance. Therefore, the main process is to extract relatively pure pixels from image $\mathbf{Y}$ as input. Although various strategies have been proposed for endmember bundles construction, such as [36], [37], the parameters of these techniques need to be carefully adjusted and do not sufficiently exploit the spatial information in the scene. Here, we develop a new and effective method for extracting the endmember bundles.

The illustration of the region-based endmember bundles extraction approach is presented in Fig. 3. The basic idea is to find a reasonably accurate estimation of pure pixels $\mathbf{M}_0 \in \mathbb{R}^{B \times R}$ [which is illustrated in Fig. 3(d)] from the subsets of the original HSI. Then, the endmember bundles can be clustered from the image pixels under appropriate distance metrics. According to the researches [38]–[40], the spectral features of adjacent pixels are highly correlated and always share the same statistical properties, which means that spectrally pure signatures are more likely to appear in spatially homogeneous areas. Hence, many existing methods adopt image segmentation to generate spatial groups and further characterizing spectral purity. Superpixel segmentation is currently the mainstream method of image segmentation. Compared with the traditional fixed size and shape of window-based methods,

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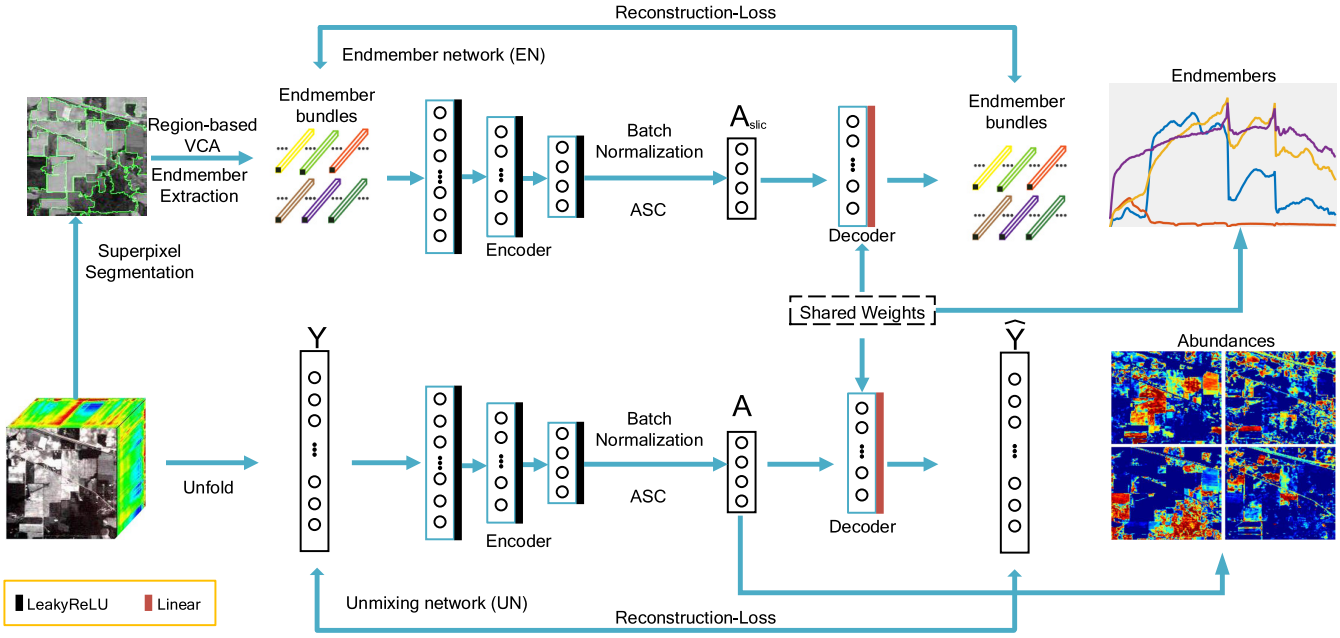


Fig. 2. Flowchart of TANet that consists of a two-stream architecture. The first stream denotes the EN for mapping the endmember bundles to their abundances $\mathbf{A}_{\text{slic}}$, and superpixel segmentation is taken to cut the HSI into homogeneous spatial regions as preprocessing. The second stream denotes the UN for taking untied-weighted autoencoder to minimize reconstruction errors. The endmember signatures are obtained from the weight matrix of two decoders, and the estimated abundances $\mathbf{A}$ are obtained from the latent vector of UN.

the separated regions can be accustomed based on different spatial structures. Especially in terms of the noise and outliers, it can exhibit more comparative performance and significantly improve calculation efficiency. In this article, we adopt the simple linear iterative clustering (SLIC) [41] as the image preprocessing. Inspired by [42], the approach is to run the endmember extraction algorithm on randomly selected subsets of the original HSI. After the endmembers are extracted from every subset, the endmembers bundles can be clustered for each ground component with the ED as the similarity measure.

Here, we adopt a more robust and easier strategy to select pure pixels from an HSI. After adapting the SLIC based superpixel segmentation, the image is over segmented into P superpixels. Then, P mean spectra can be obtained by averaging each superpixel. Finally, the reference signature $\mathbf{M}_0$ can be obtained by adopting a pure-pixel-based endmember extraction (e.g., VCA) running within the P mean spectra. We experimentally find the region-based initialization that taking the mean spectra of each superpixel as input is more robust to noise and outliers. Next, we should find suitable distance metrics to measure the similarity with the reference signature $\mathbf{M}_0$. In most of the strategies, ED and spectral angle distance (SAD) are often adopted to measure the spatial distance and the spectral distance, respectively. After different experiments, we find that the spectral information divergence (SID)-SAD method [43] can better capture the variability and similarity between different pixels in the image. Hence, in our approach, SID-SAD and ED are adopted to measure the spectral distance and the spatial distance, respectively. The mathematical form can be expressed as

$$\begin{aligned}\mathbf{D}_{\text{spectral}} &= \text{SID}(\text{x}_i, y_c) \times \tan(\text{SAD}(\text{x}_i, y_c)) \\ \mathbf{D}_{\text{spatial}} &= \sqrt{(ix - cx)^2 + (iy - cy)^2}\end{aligned} \tag{5}$$

where $\mathbf{D}_{\text{spectral}}$ and $\mathbf{D}_{\text{spatial}}$ denote the spectral and spatial distances, respectively. x_i denotes the ith reference signature of $\mathbf{M}_0$, and y_c denotes the cth candidate pixels from HSI. (ix, iy) and (cx, cy) denote the spatial coordinates of the signature and candidate pixels, respectively. SID and SAD measurements are defined as

$$\begin{aligned}\text{SID}(x_i, y_c) &= D(x_i||y_c) + D(y_c||x_i) \\ &= \sum_{k=1}^{B} p_{\text{ik}} \log\left(\frac{p_{\text{ik}}}{p_{\text{ck}}}\right) + \sum_{k=1}^{B} p_{\text{ck}} \log\left(\frac{p_{\text{ck}}}{p_{\text{ik}}}\right) \\ &= \sum_{k=1}^{B} (p_{\text{ik}} - p_{\text{ck}}) \log\left(\frac{p_{\text{ik}}}{p_{\text{ck}}}\right)\end{aligned} \tag{6}$$

$$\text{SAD}(x_i, y_c) = \cos^{-1}\left(\frac{< x_i, y_c >}{||x_i||||y_c||}\right) \tag{7}$$

where $p_{\text{ik}} = (x_{\text{ik}} / \sum_{k=1}^{B} x_{\text{ik}})$, $p_{\text{ck}} = (x_{\text{ck}} / \sum_{k=1}^{B} x_{\text{ck}})$, and x_{ik} denotes the kth band of pixel x_i.

Besides, in order to integrate the spatial and spectral priors simultaneously into the SU process, the additional parameter λ is added to balance and weight the relative importance between the spatial and spectral similarities. The final spatial–spectral distance is defined as

$$\mathbf{D} = (1 - \lambda) \times \mathbf{D}_{\text{spectral}} + \lambda \times \mathbf{D}_{\text{spatial}}. \tag{8}$$

B. Autoencoder Framework of UN

As illustrated from Fig. 2, the EN and UN are designed to share the same weights of the decoder. In addition, the specific layer types of the two networks are also the same to make it possible to take the endmember properties fully into consideration during the process of unmixing.

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TABLE I
LAYER TYPE AND ACTIVATION FUNCTION USED IN THE ENCODER

Layer	Layer Type	Activation function	Units
1	Hidden	LeakyReLU	9R
2	Hidden	LeakyReLU	6R
3	Hidden	LeakyReLU	3R
4	Hidden	LeakyReLU	R
5	Batch Normalization	Utility	R
6	Soft Thresholding	ReLU	R
7	ASC	Utility	R

The UN is designed to be a three-layer network (see the bottom of Fig. 2), where the encoder is used for dimensional reduction purposes ($\mathcal{G}_E : \mathbb{R}^{B\times 1} \to \mathbb{R}^{R\times 1}$), and the decoder is used to reconstruct the data from the learned low-dimensional representation ($\mathcal{G}_D : \mathbb{R}^{R\times 1} \to \mathbb{R}^{B\times 1}$). The detailed layer type of the encoder is listed in Table I.

Layers 1–4 are performed to compresses the input $\mathbf{Y}$ into a low-dimensional representation, and the transformation can be mathematically expressed as

$$\boldsymbol{a}^{(l)} = f\left(\boldsymbol{W}^{(l)}\boldsymbol{a}^{(l-1)}\right) \tag{9}$$

where $\boldsymbol{a}^{(l)}$ denotes the output of the hidden layer, $\boldsymbol{a}^{(l-1)}$ denotes the input from the previous layer, and $\boldsymbol{W}^{(l)}$ denotes the weight matrix, which is a trainable parameter set. Here $f(.)$ indicates an elementwise nonlinear activation function. In our experiment, we test our proposed method with the various activation function, such as Sigmoid, ReLU, or LReLU. Because the standard ReLU function has a nonzero gradient for all inputs [44], LeakyReLU (LReLU) is considered to be a better activation function in our network. It is noteworthy that the layers in our proposed network are zero-biased designed. Since bias terms tend to produce a large negative value during the training procedure, we removed the bias terms in both encoder and decoder network. The specific adverse effects can be found in [27].

Layer 5 is a batch normalization layer parameterized by two learnable parameters γ and β, which applies a transformation that maintains the mean activation of every unit close to zero and then applies scale and shifting. It plays an important role in data whitening, which can accelerate the learning speed of neural network [45]. The function of the batch normalization layer can be mathematically expressed as

$$a_i^{(l)} = \mathrm{BN}_{\gamma,\beta}\left(a_i^{(l-1)}\right) = \gamma\,\hat{a}_i^{(l-1)} + \beta \tag{10}$$

where

$$\begin{aligned}
\hat{a}_i^{(l)} &= \frac{a_i^{(l-1)} - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} \\
\mu_{\mathcal{B}} &= \frac{1}{m}\sum_{i=1}^{m} a_i^{(l-1)} \\
\sigma_{\mathcal{B}}^2 &= \frac{1}{m}\sum_{i=1}^{m}\left(a_i^{(l-1)} - \mu_{\mathcal{B}}\right)^2
\end{aligned} \tag{11}$$

where γ and β are learnable parameters, and ϵ is a positive parameter to prevent the values input from approaching infinity.

To solve the unmixing problem, there are three additional physical constraints that need to be satisfied, i.e., the nonnegativity requirement of the endmembers [for each endmember vector m_j with size $B \times 1$, $m_{\mathrm{jk}} \geq 0$, $(k = 1, \ldots, B)$] and the ANC and ASC requirement of the abundance vectors ($\alpha_{\mathrm{nj}} \geq 0$ and $\sum_{j=1}^{R} \alpha_{\mathrm{nj}} = 1$). Hence, the nonnegative and sum-to-one constraints should be incorporated to make the obtained results meaningful. Layer 6 is designed to be a dynamic thresholding layer. Here, we set θ_α as the activation function, which can be mathematically expressed as

$$\boldsymbol{a}^{(l)} = \theta_{\boldsymbol{\alpha}}\left(\boldsymbol{a}^{(l-1)}\right) = \max\left(0, \boldsymbol{a}^{(l)} - \boldsymbol{\sigma}\right) \tag{12}$$

where $\boldsymbol{\sigma}$ is an $R \times 1$ vector. The ReLU activation function is used to automatically satisfy the nonnegativity of abundance. In addition, in order to seek the sparsity of abundance maps, we choose this dynamic threshold $\boldsymbol{\sigma}$, which is a learnable parameter at the end unit in each layer to exploit the sparsity property. We have tried taking l_1 or l_{21} norm on the abundance maps to seek the sparsity constraints. However, the experimental results show that a learnable adjustment parameter is more robust than the regularization term added to the objective function.

Layer 7 is an ASC layer that is designed to ensure the abundance meets the sum-to-one constraints. We set H as the activation function. The function can be mathematically expressed as

$$a_i^{(l)} = \mathrm{H}a^{(l-1)} = \frac{a_i^{(l-1)}}{\sum_{j=1}^{R} a_j^{(l-1)}}. \tag{13}$$

Hence, combining all the previous layers, the transformation of the encoder of the UN can be written as

$$\mathcal{G}_E(\boldsymbol{y}) = \mathrm{H}\theta_{\boldsymbol{\alpha}}\left(\mathrm{BN}\left(\boldsymbol{a}^{(5)}\right)\right) \tag{14}$$

where $a^{(5)}$ denotes the output of the previous five hidden layers.

According to the LMM, the observed spectra are linearly combined by the endmembers and their respective abundance coefficients. Hence, for the decoder of UN, the activation function should also be linear. The weight matrix of the decoder has to be a $B \times R$ matrix, and the transformation of the decoder can be mathematically expressed as

$$\mathcal{G}_D = \mathbf{W}_D\mathcal{G}_E(\boldsymbol{y}) \tag{15}$$

where the columns of $\mathbf{W}_D$ exactly denote the endmembers, and $\mathcal{G}_E(\boldsymbol{y})$ denotes the output of the encoder.

C. Autoencoder Framework of EN

In our network, the detailed layer of EN is totally the same as the autoencoder-based UN. Hence, the specific layer type can be referred to Table I. For the most existing autoencoder-based approaches, the weight matrix of the decoder is directly regarded as the output of the endmember signature and the traditional training is based on end-to-end reconstruction error criterion. Hence, their ability to handle

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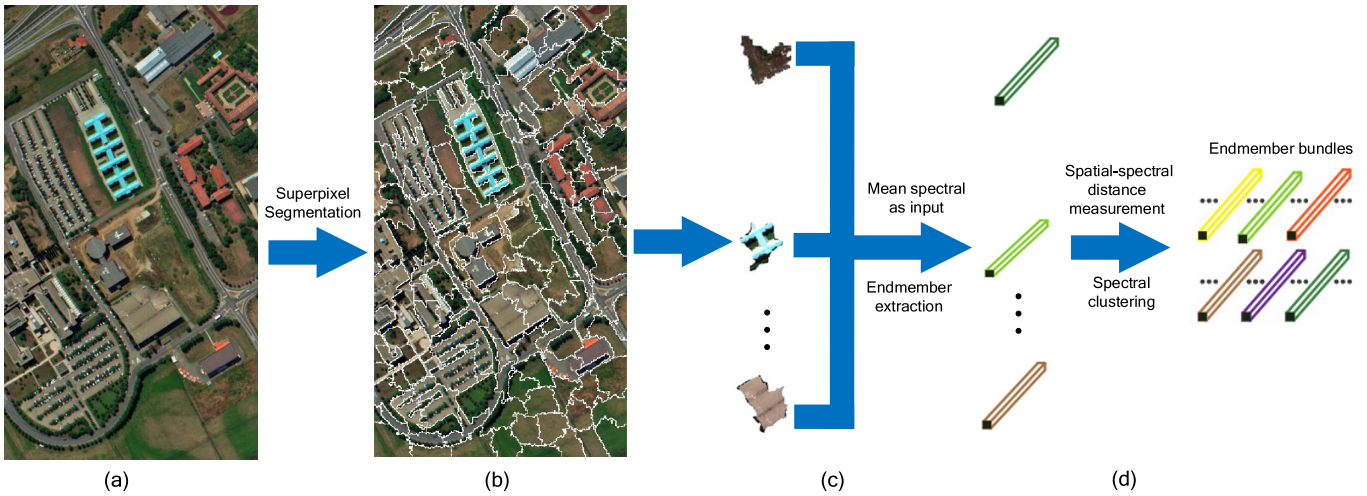


Fig. 3. Illustration of endmember bundles extraction. The image is first over segmented by the superpixel segmentation strategy as processing. Then, taking the mean spectra of each superpixel as input, the endmember bundles can be clustered for each ground component by the spatial–spectral distance measurement. (a) Pavia RGB image. (b) Superpixel map. (c) Superpixels. (d) Reference signature ($\mathbf{M}_0$).

spectral variability and extract physically meaningful endmembers remains limited. When such algorithms involve more complex scenarios, the noise and initialization conditions will greatly affect the performance of unmixing. This limitation exists more or less in the autoencoder-based approaches. In solving the above problems, DAEN [29] adopts stacked autoencoders (SAEs) to learn spectral signatures, so as to generate a good initialization for the unmixing process. Nevertheless, SAEs need extremely high computation when the image is relatively large. uDAS [28] incorporates the denoising capacity into the network and adopts the l_{21} norm to enhance unmixing performance. Hence, how to take the endmember properties fully into consideration and solve the problem of unstable initialization during the process of unmixing can greatly improve the unmixing performance. In terms of the training procedure of EN, mapping the pseudopure pixels (endmember bundles) to their abundances can be regarded as the pretraining of the network. Since the pseudopure pixels extracted by region-based contain much fewer mixed pixels, the network will be able to learn the property of endmember more effectively. Furthermore, in terms of the procedure of the UN, the parameters shared with the UN can regard as selecting every endmember combination from the endmember bundles with the smallest error, which can correct the network toward a more accurate and interpretable unmixing solution. From this perspective, this process of training is similar to the spirit of the traditional endmember set-based methods. However, the set-based methods both face an extreme searching efficiency problem when the endmember collections are large. The deep learning-based methods can solve such problems more efficiently due to their superior learning ability.

D. Objective Functions

In this article, the SAD is adopted as a measure matrix in the objective function for the neural network. Note that some other autoencoder-based methods take root mean square error (RMSE) as the reconstruction error. Nevertheless, RMSE is sensitive to the scale of the spectrum, which may lead to higher reconstruction error when the training data just scaled based on its absolute magnitude. We experimentally find that taking the scale-invariant SAD as an objective function can provide much better unmixing performances. Although such scale-invariant property could potentially lead to an incorrect scale of estimated endmembers, as long as the ASC and ANC of abundances are enforced in the neural network, the relative scale of the endmember spectra is not affected. According to the two-stream network flowchart, the loss function consists of two parts: the reconstruction error of the original data Loss_{UN} and the reconstruction error of the pseudopure pixels Loss_{EN}. Furthermore, in order to let the autoencoder converge faster, we add a regularization term reg to the loss function. The final objective functions can be mathematically expressed as

$$\begin{aligned}\text{Loss} &= \text{Loss}_{\text{UN}} + \text{Loss}_{\text{EN}} + \mu\,\text{reg} \\ &= \frac{1}{N}\left(\sum_{i}^{N}\text{SAD}(\mathbf{y}_i, \hat{\mathbf{y}}_i)\right) + \frac{1}{K}\left(\sum_{k}^{K}\text{SAD}(\mathbf{m}_k, \hat{\mathbf{m}}_k)\right) \\ &\quad + \mu\frac{1}{R}\left(\sum_{j}^{R}\left(\text{SAD}\left(\mathbf{M}_j^{est}, \mathbf{M}_{0j}\right)\right)\right) \end{aligned} \tag{16}$$

where $\mathbf{y}_i$ and $\hat{\mathbf{y}}_i$ denote the input pixel data and the reconstructed pixel data, respectively. K denotes the number of candidate pure pixels extracted to construct endmember bundles. $\mathbf{m}_k$ and $\hat{\mathbf{m}}_k$ denote the extracted endmembers bundles and reconstructed data, respectively. $\mathbf{M}_j^{est}$ denotes the estimated endmembers signature (the weight matrix of decoder), and $\mathbf{M}_{0j}$ denotes the jth of the reference signature $\mathbf{M}_0$ obtained by the region-based endmember extraction.

IV. Experimental Results

In this section, we compare the proposed method with some state-of-the-art unmixing methods, including both synthetic and real data. The representative algorithms that we choose to compare include spatial compositional model (SCM) [46],

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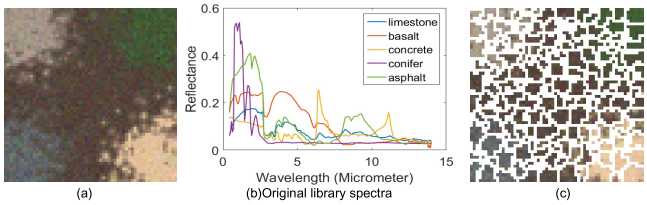


Fig. 4. (a) Synthetic color images (488, 556, and 693 nm). (b) Mean spectral extracted from the ASTER library. (c) Superpixel map used for the synthetic dataset, which is adopted to extract the endmember bundles, and the number of superpixels is to 140.

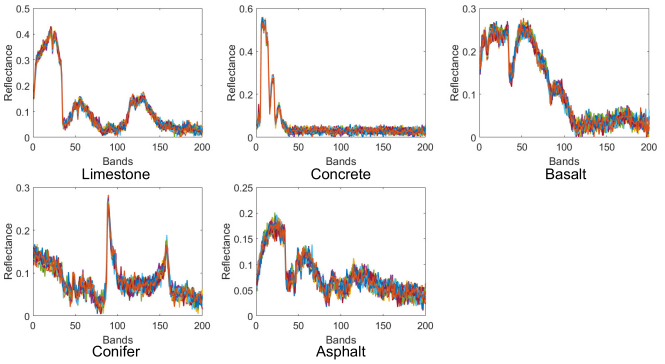


Fig. 5. Endmember bundles extracted by region-based method for the synthetic dataset, and each set contains 30 candidate pure pixels.

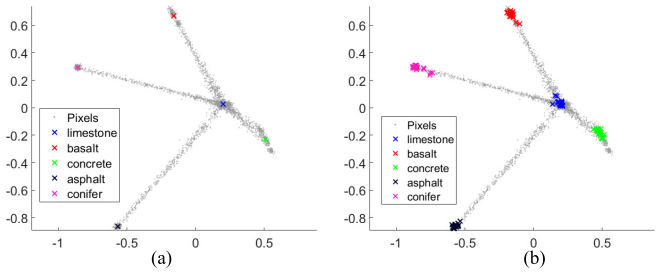


Fig. 6. Scatter plot of the synthetic dataset, where the gray dots denote the projected pixels by PCA. (a) Endmember scatter plot of the ground truth, where the darkened dots with a color represent the endmember ground truth for a material. (b) Scatter plot of the endmember bundles extracted by region-based method, where the darkened dots with a color represent the corresponding obtained endmember bundles for a material; each set contains 30 candidate pure pixels.

fully constrained least square (FCLS) [47], Gaussian mixture model (GMM) [48], spatial group sparsity regularized NMF (SGSNMF) [49], DAEN [29], and uDAS [28], where SCM takes full use of the local spatial information and the abundances are subjected to smooth and sparsity constraints to promote performance. FCLS is a traditional unmixing method of solving least square problems. GMM assumes that endmembers can be represented as a mixture of Gaussian distribution, which is a representative algorithm taking the endmember variability into consideration. DAEN and uDAS are recently proposed autoencoder-based approaches and achieving satisfactory unmixing performance. Note that DAEN adopts SAEs to generate endmembers initialization for the unmixing process. Nevertheless, we find that the SAE network needs extremely high computation. Note that, in the following experiments, the number of endmembers is assumed preknowing, which are automatically estimated by HySime [50]. For fair comparisons, the unmixing performance of the above algorithms is reported taken VCA as initialization conditions in the following experiments. In detail, the parameter of activation LeakyReLU is set to 0.2, and the optimization of training loss is ADAM with $10e-4$.

A. Synthetic Datasets

We use simulated hyperspectral data to evaluate the effectiveness of our newly proposed TANet. The dataset is created with size 60×60 pixels and constructed from endmember classes: limestone, basalt, concrete, and asphalt, whose spectra are randomly selected from the ASTER spectral library [51]. The spectra range is determined from 0.4 to 14 μm. In addition, in order to better simulate the endmember variability existing in the real HSIs, slight changes are added based on the original spectral. The asphalt material is selected as the image background; the other four materials are randomly placed in the corner. The specific generation step of endmembers follows [48]. In order to compare and analyze the robustness of the proposed method to noise, the additive noises are added to generate the mixed pixels, which are sampled from the Gaussian distribution $\mathcal{N}(\mathbf{n}_n \mid 0, \mathbf{D})$ ($\mathbf{D} = \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_B^2) \in \mathbb{R}^{B\times B}$). The corresponding abundance of each pixel is sampled from the Dirichlet distribution. Fig. 4(a) and (b) shows the resulting color images and the corresponding reference endmember signatures, respectively.

In addition to the qualitative comparison via visual inspection, five metrics are used to measure the accuracy of the unmixing results, including SAD, RMSE, SID, abundance angle distance (AAD), and abundance information divergence (AID) [12]. The mathematical form can be expressed as

$$\begin{cases} \text{SAD} = \cos^{-1}\left(\dfrac{<\mathbf{m}_i, \hat{\mathbf{m}}_i>}{||\mathbf{m}_i||||\hat{\mathbf{m}}_i||}\right) \\ \text{SID} = D(\hat{\mathbf{m}}_i||\mathbf{m}_i) + D(\mathbf{m}_i||\hat{\mathbf{m}}_i) \\ \text{RMSE} = \dfrac{1}{N}\displaystyle\sum_{j=1}^{n}\sqrt{\|\mathbf{a}_j - \hat{\mathbf{a}}_j\|_2^2} \\ \text{AAD} = \cos^{-1}\left(\dfrac{<\mathbf{a}_j, \hat{\mathbf{a}}_j>}{\|\mathbf{a}_j\|\|\hat{\mathbf{a}}_j\|}\right) \\ \text{AID} = D\big(\hat{\mathbf{a}}_j||\mathbf{a}_j\big) + D\big(\mathbf{a}_j||\hat{\mathbf{a}}_j\big) \end{cases} \tag{17}$$

where $\mathbf{m}_i$ and $\hat{\mathbf{m}}_i$ denote the extracted endmember and the library spectrum, and $\mathbf{a}_j$ and $\hat{\mathbf{a}}_j$ denote the corresponding abundance ground truth and the estimated values, respectively. The specific measurement of SID are defined in (6). For those five measurement metrics, the lower the value of the matrix, the better the performance of the algorithm.

1) Parameter Setting: In the implementation of the proposed method, there are four parameters that should be adjusted: the spatial–spectral balanced term λ, the number of superpixels P, the number of candidate pure pixels K extracted for endmember bundles, and the regularization term μ. The first parameter λ weights the relative importance between the spatial and spectral similarities, which can

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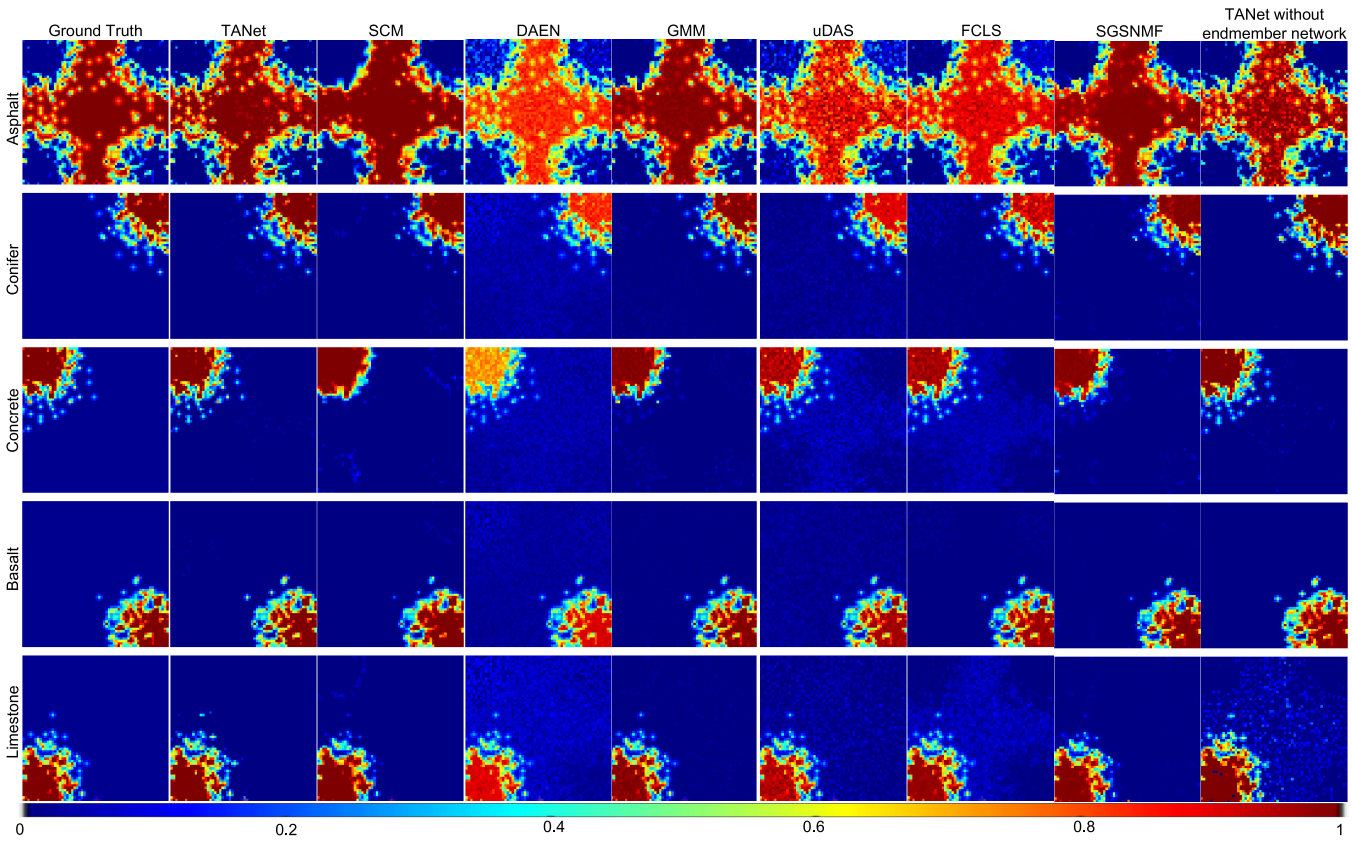


Fig. 7. Estimated abundance maps for the synthetic dataset with Gaussian noise set to SNR = 20 dB. (From left to right) Abundance maps comparisons of ground truth, TANet, SCM, DAEN, GMM, uDAS, FCLS, SGSNMF, and TANet without EN. (From top to bottom) Corresponding endmembers of asphalt, conifer, concrete, basalt, and limestone.

be set in the range [0.1, 1]. Considering that the spectral information occupies a larger proportion in measuring the similarity, the optional value of λ is suggested to be less than 0.5. In our synthetic and real datasets, λ is both set to 0.3. The second parameter P is utilized to control the total number of spatial groups. The image segmentation is used to generate spatially homogeneous areas, which is mainly helping to find a reasonably accurate initialization of extracted endmember bundles. Thus, the parameter of superpixels P does not need to be carefully adjusted but only satisfies the requirement of oversegmentation. In this experiment, the superpixels P are set to 140, and the corresponding decomposition of the synthetic image is illustrated in Fig. 4(c). For the third parameter, K is set to 150 for the synthetic dataset. Hence, 30 pixels closest to the reference matrix $\mathbf{M}_0$ in each endmember set are selected to construct the endmember bundles. The resulting endmember bundles in this experiment are illustrated in Fig. 5. The corresponding scatter plot of the endmember bundles is illustrated in Fig. 6.

2) Comparison to State-of-the-Art Unmixing Algorithms: In this experiment, all algorithms take VCA [8] as initialization for endmember extraction. The number of endmembers is assumed as $R = 5$. Fig. 7 shows the abundance comparison with the synthetic dataset when the SNR is set at 20 dB. Comparing the ground truth (the first row of Fig. 7), we can find that the abundance map of asphalt is generated at the center, while the regions of the other four endmembers' abundance maps (conifer, concrete, basalt, and limestone) are randomly located at four corners. Hence, our proposed method can provide a more accurate estimation since the abundance maps look relatively clean and less cluttered. The corresponding estimated endmember comparisons are illustrated in Fig. 8. Comparing the endmember results, we can clearly find that, when the endmembers are in the same initial condition, DAEN and FCLS show the insufficiency in this case. The quantitative performance analysis of these algorithms is shown in the second row of Table II (SNR = 20 dB), which shows the effectiveness of our proposed algorithm.

3) Analysis of Robustness to Noise: We carried out this experiment to compare and analyze the robustness of the algorithms. In this experiment, the synthetic dataset was polluted by the white Gaussian noise with different levels, where the SNR is varied from 10 to 30 dB with an interval of 10 dB. The quantitative analysis of these algorithms is shown in Table II. Note that all the algorithms are performed 10 times on each dataset, and the average value is reported in Table II. In general, all the algorithms exhibit decreased unmixing accuracy as the noise level increases. We can find that the proposed method exhibits much better robustness in the five measurement metrics. In detail, for two autoencoder-based approaches, we find that DAEN performs quite well when the noise level is set SNR = 30 dB. Compared with the higher noise case (SNR = 10, 20 dB), the performance of unmixing is greatly improved. The results of instability are also related to

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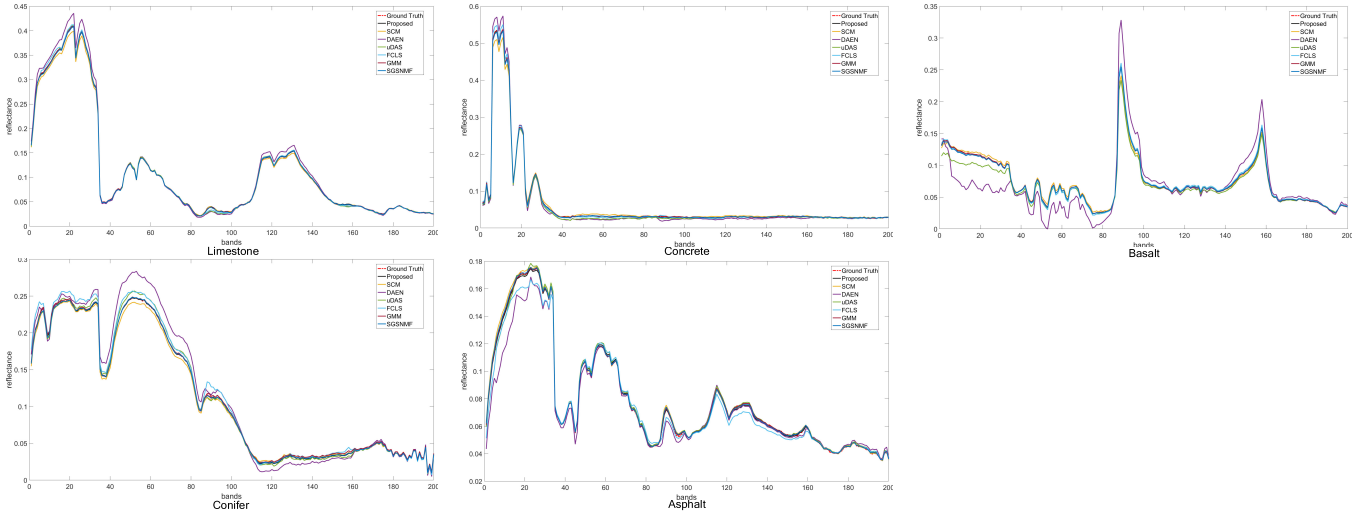


Fig. 8. Endmember estimation results for the synthetic dataset.

TABLE II
UNMIXING EVALUATION ON SYNTHETIC DATASET WITH GAUSSIAN NOISE FROM SNR = 10–30 dB

SNR=10dB	TANet	SCM	DAEN	GMM	uDAS	FCLS	SGSNMF
Limestone	1.80e-2/**6.54e-2**[a]	2.43e-2/0.147	9.45e-2/0.153	**1.38e-2**/9.42e-2	6.60e-2/0.211	0.109/0.203	1.58e-2/8.27e-2
Basalt	**1.72e-2/3.55e-2**	2.58e-2/6.58e-2	5.74e-2/6.21e-2	4.20e-2/5.46e-2	3.55e-2/5.24e-2	4.56e-2/8.35e-2	2.18e-2/3.66e-2
Concrete	2.6e-2/**3.42e-2**	5.91e-2/9.80e-2	0.2712/9.12e-2	**2.41e-2**/7.30e-2	7.14e-2/0.106	0.126/0.112	3.01e-2/5.27e-2
Conifer	**1.81e-2/1.76e-2**	5.05e-2/5.13e-2	7.40e-2/4.69e-2	3.73e-2/4.70e-2	5.72e-2/4.35e-2	4.99e-2/2.96e-2	3.59e-2/2.74e-2
Asphalt	**1.44e-2**/3.71e-2	1.75e-2/5.38e-2	2.90e-2/6.62e-2	1.72e-2/4.14e-2	3.04e-2/6.95e-2	3.04e-2/4.32e-2	1.73e-2/**3.18e-2**
Mean	**1.88e-2**/3.80e-2	3.55e-2/8.32e-2	0.105/8.40e-2	2.68e-2/6.20e-2	5.21e-2/9.65e-2	7.22e-2/9.42e-2	2.42e-2/4.62e-2
AAD	**6.02**[b]	12.58	10.47	9.47	13.24	13.56	7.25
AID	0.94[b]	2.43	2.96	1.60	3.38	3.39	**0.43**
SID	**1.2e-3**[b]	5.9e-3	4.41e-2	4.7e-3	8.0e-3	1.68e-2	2.9e-3

SNR=20dB	TANet	SCM	DAEN	GMM	uDAS	FCLS	SGSNMF
Limestone	1.12e-2/**3.60e-2**	1.4e-2/9.92e-2	7.18e-2/0.124	**6.9e-3** /4.83e-2	1.83e-2/0.103	4.21e-2/7.41e-2	8.4e-3/6.68e-2
Basalt	**4.1e-3**/1.86e-2	1.25e-2/3.26e-2	6.27e-2/6.11e-2	8.0e-3/**8.4e-3**	2.05e-2/3.50e-2	2.11e-2/2.87e-2	5.0e-3/2.62e-2
Concrete	**4.5e-3/1.51e-2**	3.53e-2/7.08e-2	0.293/8.93e-2	4.7e-3/4.44e-2	4.80e-2/4.24e-2	2.12e-2/3.64e-2	8.8e-3/4.48e-2
Conifer	**4.7e-3**/1.31e-2	3.47e-2/3.23e-2	5.93e-2/3.82e-2	8.2e-3/5.7e-3	2.83e-2/2.15e-2	1.68e-2/**1.17e-2**	5.7e-3/1.73e-2
Asphalt	**3.3e-3/2.33e-2**	9.8e-3/3.13e-2	2.46e-2/6.14e-2	4.0e-3/8.2e-3	1.28e-2/3.18e-2	1.02e-2/3.92e-2	3.9e-3/2.36e-2
Mean	**5.5e-3/2.12e-2**	2.13e-2/5.33e-2	0.1023/7.48e-2	6.4e-3/2.30e-2	2.56e-2/4.68e-2	2.23e-2/3.80e-2	6.4e-3/3.57e-2
AAD	**3.30**	8.28	8.87	4.3056	5.90	5.09	5.67
AID	0.49	1.15	2.58	**0.30**	1.55	1.22	0.43
SID	**1.00e-4**	1.60e-3	6.55e-2	2.26e-4	2.40e-3	1.30e-3	1.12e-4

SNR=30dB	TANet	SCM	DAEN	GMM	uDAS	FCLS	SGSNMF
Limestone	**1.7e-3/2.68e-2**	1.07e-2/7.14e-2	2.11e-2/4.29e-2	3.7e-3/4.12e-2	2.89e-2/9.08e-2	3.42e-2/6.10e-2	5.3e-3/7.11e-2
Basalt	**1.7e-3**/8.4e-3	9.5e-3/2.06e-2	1.68e-2/1.79e-2	2.0e-3/**3.2e-3**	1.50e-2/2.88e-2	2.11e-2/2.71e-2	4.4e-3/2.83e-2
Concrete	**2.5e-3/1.65e-2**	2.45e-2/5.09e-2	3.34e-2/2.24e-2	3.6e-3/4.06e-2	4.54e-2/3.47e-2	5.33e-2/3.02e-2	6.9e-3/4.36e-2
Conifer	**1.8e-3**/1.03e-2	2.71e-2/2.21e-2	2.71e-2/1.55e-2	4.1e-3/**2.1e-3**	1.53e-2/1.52e-2	1.48e-2/1.63e-2	7.4e-3/2.06e-2
Asphalt	**1.2e-3**/1.32e-2	7.5e-3/2.13e-2	1.17e-2/1.79e-2	1.7e-3/**2.4e-3**	1.23e-2/3.96e-2	1.12e-2/2.71e-2	4.2e-3/2.60e-2
Mean	**1.80e-3/1.51e-2**	1.59e-2/3.72e-2	2.20e-2/2.33e-2	3.0e-3/1.79e-2	2.34e-2/4.18e-2	2.69e-2/3.23e-2	5.60e-3/3.79e-2
AAD	**2.32**	5.84	2.56	3.78	5.29	4.08	5.89
AID	**0.29**	0.62	0.747	0.107	1.36	1.01	0.47
SID	**9.92e-6**	9.94e-4	1.3e-5	5.10e-5	1.40e-3	1.6e-3	1.13e-4

[a] In this article, the numbers in "./." denote SAD for endmember errors and the RMSE for abundance errors.
[b] The number in each item (AAD, AID and SID) denote the mean errors of the above five materials (limestone, basalt, concrete, conifer, asphalt), respectively.

the fact that VCA initialization is not stable when the noise at different levels. For uDAS, since the network is incorporated with the denoising constraint, the performance varies little at the range of SNR = 20–30 dB. However, the overall unmixing results have not been satisfactory. As a traditional algorithm considering endmember variability, GMM has a favor performance in most of the cases in this experiment. Nevertheless, it should be noted GMM needs high memory requirements. When the spectral library is large, its complexity will increase exponentially and resulting in extreme searching efficiency problems, which can be explained in [38], [52], and [53]. The TANet outperforms the other approaches and achieves the best or comparable results in most cases, which proves the effectiveness of the algorithm.

4) Impact of the EN: In order to verify the advantages and effectiveness of our proposed method over the traditional autoencoder framework, in this experiment, we test the unmixing performance of TANet after removing the endmember framework and superpixel preprocessing. The noise level is also set to SNR = 20 dB, and the performance of abundance maps are demonstrated in the last column of Fig. 7. Comparing the performance of TANet, we can clearly find that many

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TABLE III

UNMIXING EVALUATION ON SYNTHETIC DATASET WITH GAUSSIAN NOISE SET TO SNR = 20 dB

SNR=20dB	TANet	TANet Without EN
Limestone	**1.12e-2/3.60e-2**	1.43e-2/0.101
Basalt	**4.1e-3/1.86e-2**	6.6e-3/4.40e-2
Concrete	**4.5e-3/1.51e-2**	6.5e-3/1.33e-2
Conifer	**4.7e-3/1.31e-2**	1.60e-2/3.64e-2
Asphalt	**3.3e-3/2.33e-2**	1.10e-2/8.71e-2
Mean	**5.5e-3/2.12e-2**	1.09e-2/5.64e-2
AAD	**3.39**	8.47
AID	**0.49**	1.50
SID	**1.00e-4**	5.03e-4

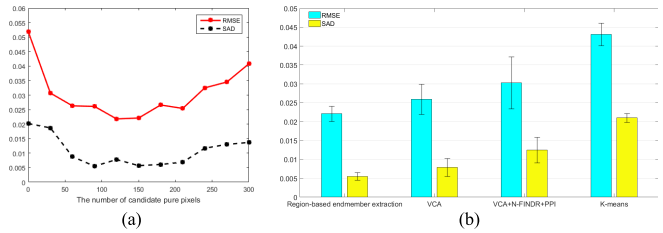


Fig. 9. (a) Performance analysis of TANet with respect to different numbers of candidate pure pixels settings when the Gaussian noise is set to SNR = 20 dB. (b) Quantitative analysis of TANet under different endmember extraction strategies; the Gaussian noise is set to SNR = 20 dB.

discrete and isolated points appear in the abundance maps of asphalt and limestone materials. The quantitative performance analysis is shown in Table III. The unmixing performance in the first column of the table has been comprehensively improved compared with that in the second column, which indicates that only taking end-to-end reconstruction error for SU cannot achieve satisfactory results. The parameters shared with EN making it possible to take the endmember properties fully into consideration during the process of unmixing further enhance the performance.

Besides, the number of candidate pure pixels K is also analyzed to evaluate the influence on the performance of the experiments. Fig. 9 illustrates the varying tendency of the unmixing performance as K increases. As can be seen from Fig. 9(a), a certain number of candidate pure pixels can greatly enhance the performance of the TANet, and stable and excellent performance is obtained when K varies from 100 to 150. The corresponding endmember scatter plot is illustrated in Fig. 6. Nevertheless, as the number of endmember bundles is set to relatively larger, the precision of the network is gradually declining. Since the parameter K in our experiments denotes the pixels that are closest to the reference matrix $\mathbf{M}_0$, a larger number of endmember bundles imply the growing number of mixed pixels, which will reduce the precision of the network. This experiment also indicates that the parameters transferred by the EN shall be effective and satisfactory. In fact, common limitations faced by TANet are the accuracy and robustness of endmember extraction algorithms. Here, we also test the performance of TANet under other three different endmember extraction strategies, including VCA, the combination of VCA, N-FINDR [54], the pixel purity index (PPI) [55], and K-means. For these three methods, the endmember bundles are obtained by repeatedly running multiple times to extract enough endmembers for training. The

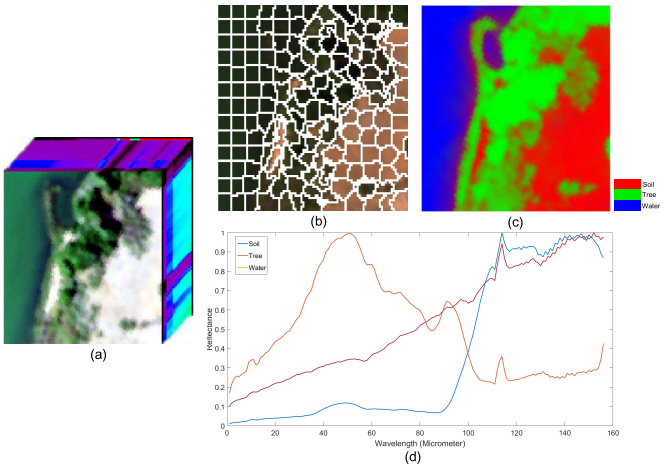


Fig. 10. (a) 3-D cube of the samson dataset. (b) Superpixel map used for the samson dataset, which is adopted to extract the endmember bundles, and the number of superpixels is 168. (c) Corresponding ground truth. (d) Endmember ground truth.

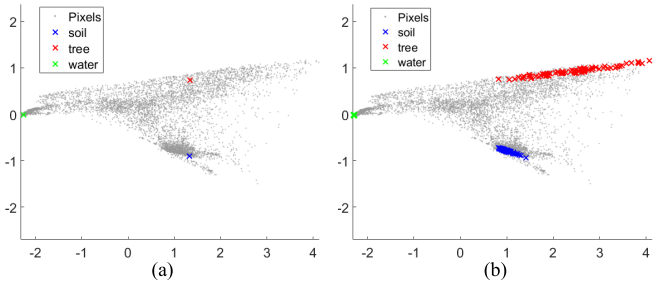


Fig. 11. Scatter plot of the samson dataset, where the dots and marks have the same meaning as in Fig. 6. (a) Endmember scatter plot of the ground truth. (b) Scatter plot of the endmember bundles extracted by the region-based method; each set contains 100 candidate pure pixels.

quantitative performance is illustrated in Fig. 9(b). We can observe that the proposed region-based endmember extraction method achieves the best and most stable unmixing results, whereas the K-means based method performs poorly because it failed to estimate the center of the endmember well under such noise conditions. VCA and the combination of VCA, N-FINDR, and PPI also obtain competitive performance compared with the corresponding quantitative results of other state of the arts in Table II (the RMSE and SAD results with SNR = 20 dB in Table II). Nevertheless, the standard deviation is large due to the instability of endmember extraction. Therefore, taking an effective method for extracting the endmember bundles can further enhance the TANet network. The competitive unmixing results also prove the effectiveness of the algorithm.

In the following experiment, in addition to those algorithms mentioned above, we report the performance of current methods for several datasets by referring directly to their reported scores since their codes are not available: $l_{1|2}$-NMF [56] and DgS-NMF [57].

B. Real-Data Experiments

1) Samson Dataset: For the real data experiments, the first real HSI was collected by the SAMSON sensor. The size

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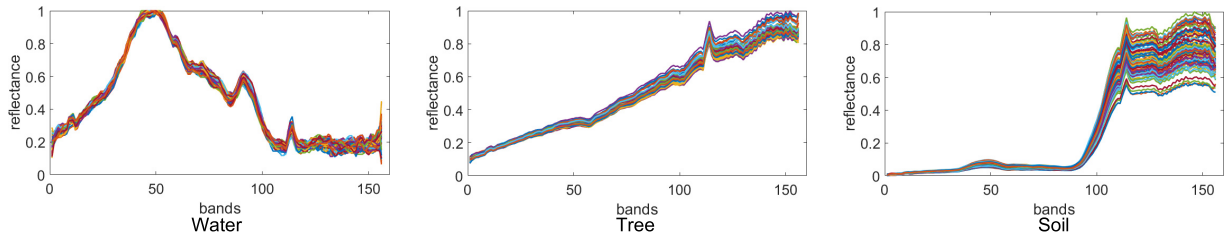


Fig. 12. Endmember bundles extracted by region-based method for the samson dataset; each set contains 100 candidate pure pixels.

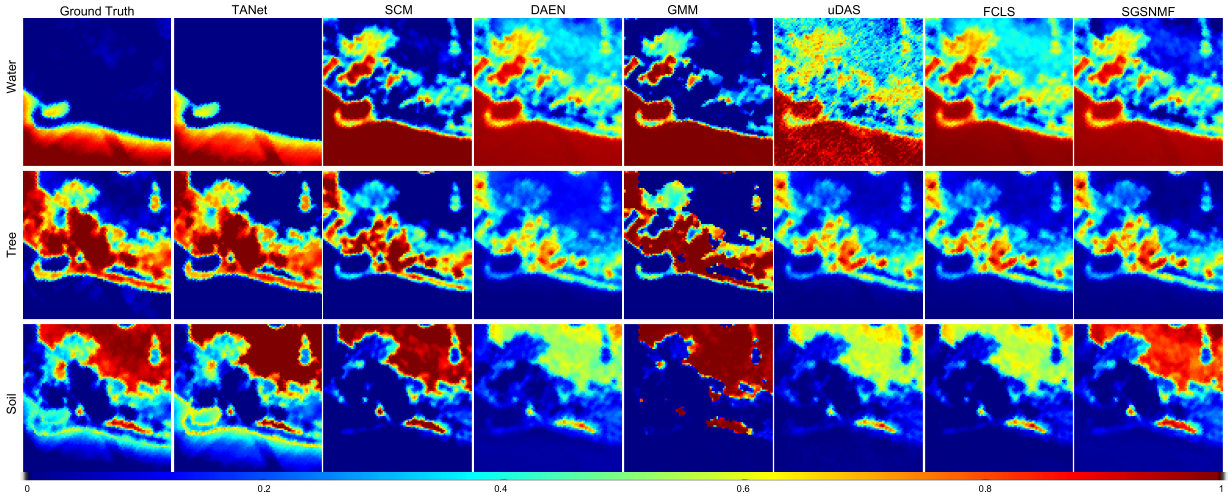


Fig. 13. Estimated abundance maps for the samson dataset. (From left to right) Abundance maps comparisons of ground truth, TANet, SCM, DAEN, NCM, uDAS, and FCLS. (From top to bottom) Corresponding endmembers of water, tree, and soil.

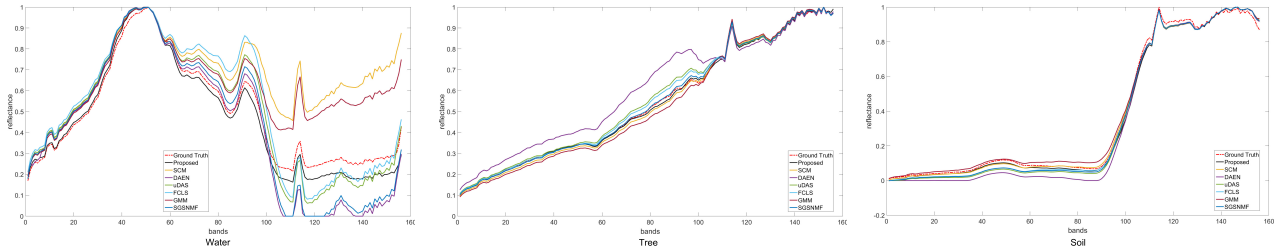


Fig. 14. Endmember estimation results for the samson dataset.

of the samson dataset is 95 × 95 pixels, and the spectra range is from 0.4 to 0.9 μm. Specifically, for validation [57], there is three material in this image observed: soil, tree, and water, in the ground-truth image. The 3-D cube image and corresponding ground truth are illustrated in Fig. 10(a) and (c). In this experiment, the spatial–spectral balanced term λ is set 0.3, and the superpixel map used is illustrated in Fig. 10(b). The corresponding scatter plot and endmember bundles are illustrated in Figs. 11 and 12, where K is set to 300 so that each set of endmembers contains 100 candidate pure pixels. The endmember regularization term μ is set to 0.1, and the batch size of the network is set to 40.

The estimated abundance maps are shown in Fig. 13. Compared to the other method, the results of TANet not only showed sparse abundances for the region but also interpreted the boundary as a combination of neighboring materials. Specifically, GMM shows insufficiency in this case. Note that GMM and other statistical learning methods that take the endmember variability into consideration both have such kind of limitations. The core of those methods is to assume that the endmember of each pixel is derived from a sample of a certain

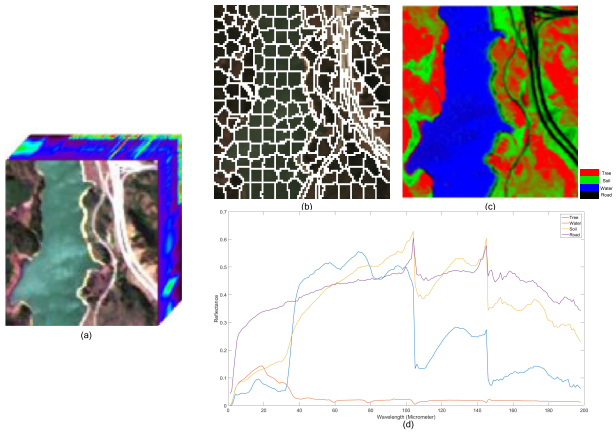


Fig. 15. (a) 3-D cube of the jasper dataset. (b) Superpixel map used for the jasper dataset, which is adopted to extract the endmember bundles, and the number of superpixels is to 208. (c) Corresponding ground truth. (d) Endmember ground truth.

probability distribution. Nevertheless, in the highly mixed scenes, there are no sufficient pure pixel samples for training and further affecting the unmixing performance. For the other

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TABLE IV
UNMIXING EVALUATION COMPARISON ON SAMSON DATASET

Material($\times 10^{-2}$)	TANet	SCM	DAEN	GMM	uDAS	FCLS	SGSNMF	$l_{1\|2}$-NMF	DgS-NMF
Soil	1.06/**6.62**	1.86/18.6	10.9/26.1	3.69/21.4	3.27/25.9	2.36/26.5	**1.01**/18.0	6.21/8.58	5.64/7.77
Tree	3.26/**4.60**	**2.97**/19.8	8.85/27.0	3.88/19.2	5.39/25.4	4.33/25.1	4.61/24.7	5.23/7.44	4.80/7.74
Water	7.95/3.31	29.6/33.3	24.8/40.3	22.9/29.1	14.1/40.8	15.1/42.3	23.1/36.0	11.97/5.65	**4.70/2.70**
Mean	**4.09/4.85**	11.1/24.0	14.9/31.1	10.2/23.2	7.57/30.7	7.24/31.3	9.58/26.2	7.80/7.19	5.05/6.07
AAD	**501.9**	2462.7	2714.9	2457.6	2667.1	2749.4	2519.0	–	–
AID	**71.2**	662.3	667.0	615.3	669.2	776.2	612.6	–	–
SID	**0.77**	69.6	69.6	4.16	4.31	3.10	44.7	–	–

TABLE V
UNMIXING EVALUATION COMPARISON ON THE JASPER DATASET

Material($\times 10^{-2}$)	TANet	SCM	DAEN	GMM	uDAS	FCLS	SGSNMF	$l_{1\|2}$-NMF	DgS-NMF
Road	**2.94/5.27**	4.44/8.60	5.26/12.3	9.48/16.0	5.99/11.9	53.3/34.7	5.26/14.1	15.1/16.2	4.66/11.7
Soil	**3.40**/5.31	7.13/11.9	14.7/18.0	13.4/26.8	11.7/13.0	22.6/43.0	14.7/11.5	4.60/5.57	4.60/**4.13**
Water	9.62/**5.58**	9.76/13.5	20.1/14.5	13.2/10.7	11.5/19.3	25.4/16.9	20.1/18.7	6.16/17.02	**5.66**/11.13
Tree	2.23/**3.58**	4.42/8.38	14.6/14.4	**2.06**/22.3	17.1/15.8	17.7/11.9	14.6/15.0	9.81/6.73	6.73/5.68
Mean	**4.55/4.93**	6.44/10.6	13.7/14.8	9.53/19.0	11.6/15.0	29.8/26.6	13.7/14.8	7.19/11.4	5.41/8.15
AAD	**800.0**	1521.0	1754.7	2594.4	1835.7	5196.2	1754.7	–	–
AID	**92.0**	217.9	335.7	452.5	330.3	1336.3	335.7	–	–
SID	2.73	**2.23**	11.9	2.95	6.87	23.4	11.9	–	–

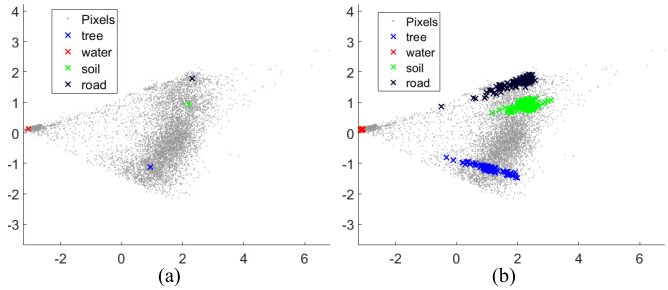


Fig. 16. Scatter plot of the jasper dataset, where the dots and marks have the same meaning as in Fig. 6. (a) Endmember scatter plot of the ground truth. (b) Scatter plot of the endmember bundles extracted by region-based method; each set contains 100 candidate pure pixels.

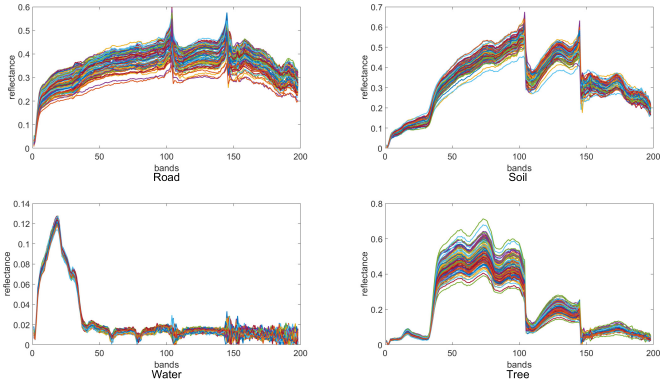


Fig. 17. Endmember bundles extracted by the region-based method for the jasper dataset; each set contains 100 candidate pure pixels.

two autoencoder-based networks, the proposed method also has a convincing improvement. The estimated endmember comparisons are shown in Fig. 14. Table IV demonstrates the obtained quantitative results for the considered methods.

2) Jasper Dataset: The second real HSI is captured by the AVIRIS sensor. The subimage that we used is 100 × 100 pixels of the original data. After removing the channels 1–3, 108–112, 154–166, and 220–224 (due to dense water vapor absorption and atmospheric effects), the region remains 198 channels, and the spectra are ranging from 0.4 to 2.5 μm. There is four materials in this image are observed: road, soil, water, and tree. The ground truths of abundance and endmember are shown in Fig. 15(c) and (d). In this experiment, the spatial–spectral balanced terms λ is set 0.3, and the superpixel map used is illustrated in Fig. 15(b). The corresponding scatter plot and endmember bundles are illustrated in Fig. 16 and Fig. 17, where K is set to 400 so that each set of endmembers contains 100 candidate pure pixels. The endmember regularization term μ is set to 1; the batch size of the network is set to 40.

The estimated abundance maps and endmember estimation results comparison are shown in Figs. 18 and 19. As illustrated in abundance maps, TANet provides a relatively accurate estimation, which has better performance results in both highly mixed region (soil) and homogeneous region (water). Table V presents the obtained quantitative values from all the considered methods. From this table, we can further prove that the proposed TANet obtains better and competitive results compared with other methods.

V. CONCLUSION

In this article, we introduce a new unsupervised hyperspectral unmixing approach that is based on a two-stream autoencoder network. The TANet mainly consists of two networks: an EN and a UN. Compared with other learning-based algorithms, the parameters shared with the same weight make it possible to transfer the potentially intrinsic properties of the endmembers into consideration. Nevertheless, most autoencoder-based methods treat the weight matrix of the decoder directly as the obtained endmember signatures; their ability to extract physically meaningful endmembers remains limited. The additional endmember bundle training error can exhibit a more accurate and interpretable unmixing

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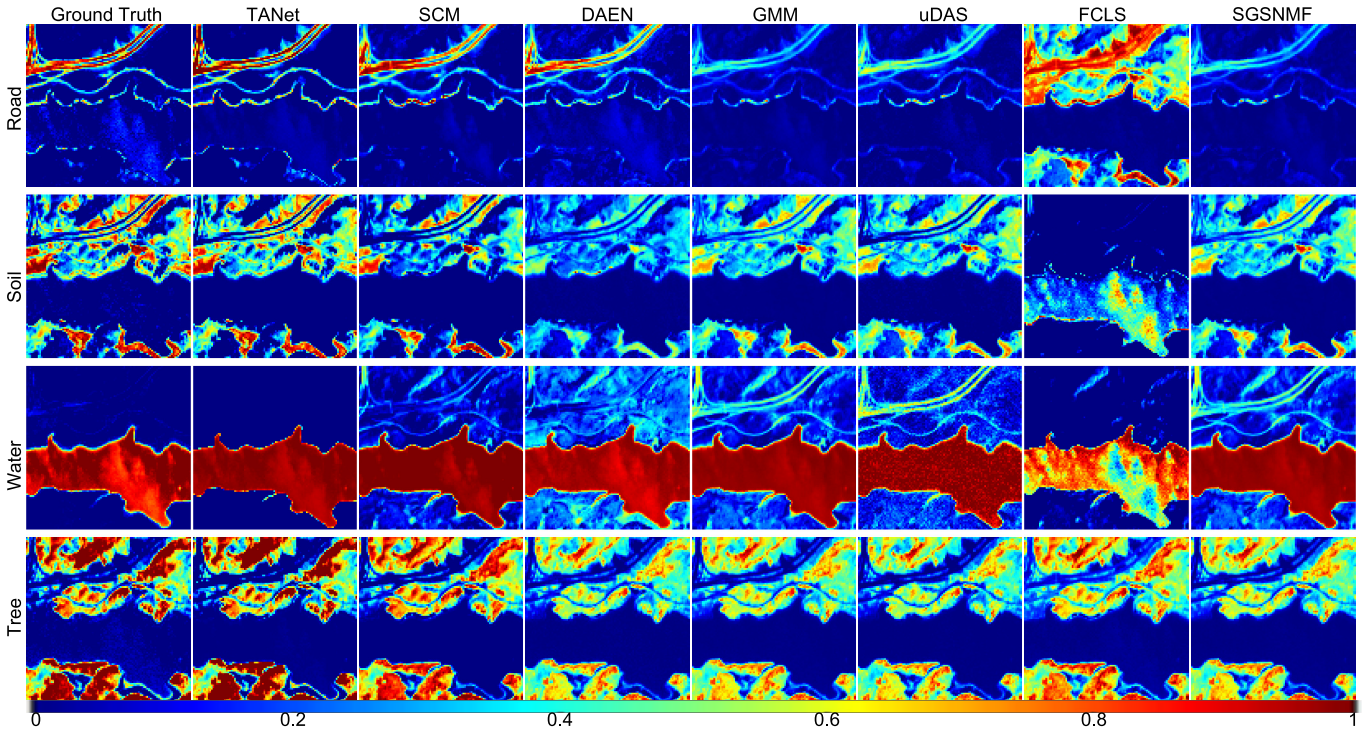


Fig. 18. Estimated abundance maps for the jasper dataset. (From left to right) Abundance maps comparisons of ground truth, TANet, SCM, DAEN, NCM, uDAS, and FCLS. (From top to bottom) Corresponding endmembers of road, soil, water, and tree.

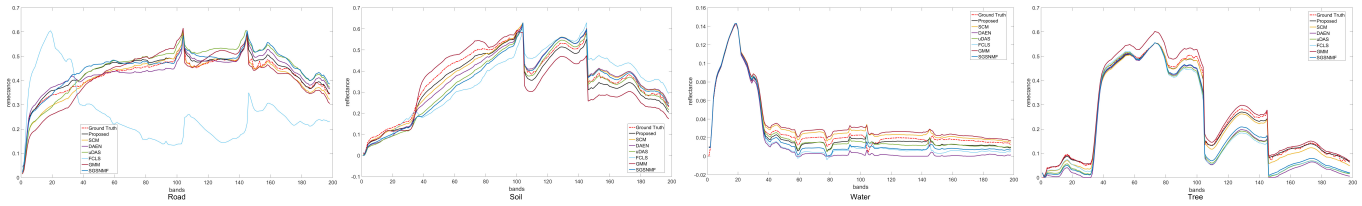


Fig. 19. Endmember estimation results for the jasper dataset.

performance. As the proposed TANet is based on the LMM, future works will develop a more general network to nonlinear unmixing scenarios. In addition, since the endmember bundles have the ability to handle spectral variability, we will also further explore the unmixing performance of TANet in dealing with the endmember uncertainty task.

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